**Voice Onset Time merger and development of tonal contrast in Seoul Korean stops: a corpus study**

Yoonjung Kang[a,b,*],

*Corresponding author: yoonjung.kang@utoronto.ca, Tel: +1 416 287 7172, Fax: +1 416 287 7176
[a] 1265 Military Trail, University of Toronto Scarborough, Centre for French and Linguistics, Toronto, Ontario, M1C 1A4, Canada
[b] 100 St. George Street, University of Toronto, Department of Linguistics, Toronto, Ontario, M5S 3G3, Canada

**Abstract**
This paper is an apparent-time study of sound change in a three-way laryngeal contrast of Korean stops. The data are drawn from a read speech corpus distributed by the National Institute of the Korean Language. Voice Onset Time (VOT) of sentence initial stops and f0 of the first two vowels of the sentence produced by 117 Seoul Korean speakers were measured to determine how VOT and f0 realization of stop categories vary by speakers' age (range: 19-71) and gender. This is the first large-scale study of its kind based on data gathered from Seoul residents. The results replicate previous findings that the VOT values of aspirated stops are shortening in younger speakers' speech with the VOT difference between aspirated and lenis stops reducing accordingly and that this change is more advanced in female speech than in male speech. The novel finding of the study is that there is a trend of enhanced f0 distinction between aspirated and lenis stops, whereby the f0 distinction is amplified and extends further into the phrase in the speech of younger compared to older speakers and in the speech of female compared to male speakers. The results show that the establishment of tonal contrast and the loss of VOT distinction are taking place in tandem rather than in separate stages. The result also indicates that f0 enhancement as sound change is mediated by structural categories, namely all [+spread glottis] sounds including /h/ rather than narrowly targeting threatened segmental contrast in stop consonants.

**Keywords**
Korean, Tonogenesis, VOT, F0, Age, Gender

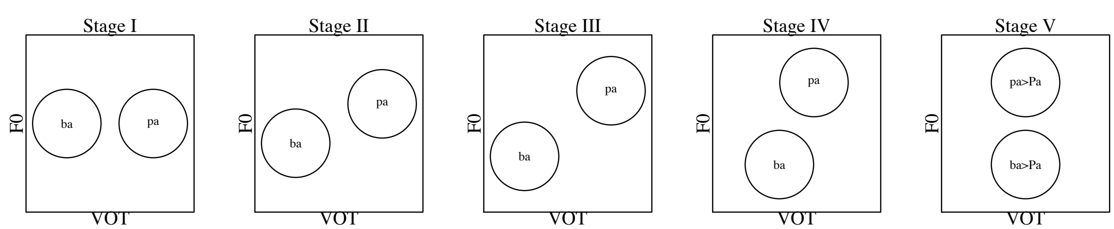
**Voice Onset time merger and development of tonal contrast in Seoul Korean stops: a Corpus study**

## 1.0 Introduction

Korean has a typologically unusual three-way contrast of voiceless stops among aspirated stops /$p^h$,$t^h$,$k^h$/, lenis stops /p, t, k/, and fortis stops /p',t',k'/. Previous studies show that the contrast is signalled by a combination of acoustic cues including voice onset time (VOT) of the stop and fundamental frequency (f0) and the amplitude difference between the first and the second harmonics (H1-H2) of the following vowel; aspirated stops have the longest VOT values, fortis stops have the shortest values, and lenis stops have intermediate values; f0 on the following vowel is higher for aspirated and fortis stops than for lenis stops; vowels following aspirated and lenis stops have breathier voice quality than vowels following fortis stops as indicated by higher H1-H2 values ((T. H. Cho, Jun, & Ladefoged, 2002; K. H. Kang & Guion, 2008; Kim, 1965) Hardcastle, 1973; Han & Weitzman, 1970; Kang & Guion, 2008; Kim, 1965; Kim, 1994; Lisker & Abramson, 1964).

It has been suggested that the stop system of Seoul Korean is undergoing a tonogenesis-like sound change (Kim, 2000; Silva, 2006; Wright 2007). In tonogenesis, consonantal contrasts of voicing, aspiration and glottalization, or phonation contrast of breathiness and creakiness give rise to and eventually become replaced by tonal contrasts (Haudricourt 1971; Hombert, 1978; Hombert, Ohala, & Ewan, 1979; Kingston, 2011; Maran, 1973; Matisoff, 1973; Thurgood 2002). In this change, tones start out as redundant phonetic attributes of consonantal contrast then develop into a robust distinction coexisting with the original consonantal contrast, and finally become the primary contrastive feature, as the original consonantal features are lost. **Fig. 1** illustrates the stages of tone development on a post-consonantal vowel by schematically representing the distribution of two-way contrast in a two-dimensional acoustic space of VOT and f0, based on five stages of tone development proposed by Maran (1973).

**Fig. 1 Five stages of tonogenesis based on (Maran, 1973)**
(Stage 1: VOT only, no f0 distinction, Stage 2: Emergence of redundant f0 distinction, Stage 3: Full redundancy, Stage 4: Weakening of VOT distinction & further enhancement of f0 distinction, Stage 5: Loss of VOT distinction)




In Seoul Korean, the VOT values reported in studies from the 1960s and the 1970s typically show a good separation between aspirated and lenis stop categories (Lisker & Abramson, 1964; Kim, 1965; Han & Weitzman, 1970; Hardcastle, 1973; Kagaya, 1974). These early studies tend to also find differentiation of f0 between the aspirated and lenis stops, indicating that the language was around Stage II or III of tonal development. Studies from the 1990s found that the VOT values of aspirated stops shortened while those of lenis stops lengthened (T. Cho, 1996; Han, 1996; Silva, 1992) indicating that the language was moving toward Stage IV. It is observed that the consonant-induced f0 perturbation is already "phonologized" (Jun, 1996, p.111); unlike English and French, consonant-induced f0 distinction in Korean extends far beyond the initial portion of the immediately following vowel.

Specifically, Accentual Phrases (AP) that begin with a [+stiff vocal folds] segment—a fortis or aspirated stop, /h/, or coronal fricatives—are marked by HH boundary tones with H falling on the first two syllables, while phrases that begin with a [-stiff vocal folds] segment—a lenis stop or a sonorant—are marked by LH boundary tones with L falling on the initial syllable and H on the second (Jun, 1993, 1996). The realization of AP-initial boundary tones is schematically represented as in (1).

**(1) Accentual Phrase-initial boundary tones in Seoul Korean**

$_{AP}$[ σ1 σ2 ….. ]

L H if σ1-initial segment is [-stiff vocal folds], i.e., lenis stops or sonorants

H H if σ1-initial segment is [+stiff vocal folds], i.e., fortis or aspirated stops, coronal fricatives, or /h/

Finally, studies from the last decade or so found that lenis and aspirated stops overlap substantially in their VOT values while continuing to find a robust f0 difference between the two categories (Choi 2002; Kim, 2004; Silva, 2006; Wright 2007; Kang & Guion, 2008; Kong, Beckman, & Edwards, 2011; Oh, 2011; Lee & Jongman, in press). Perception studies also find that f0 is a crucial perceptual cue for stop contrast, for lenis vs. aspirated stop contrast in particular (Kim, Beddor, & Horrocks, 2002; Kim, 2004; Lee & Jongman, 2011; Kong, Beckman, & Edwards, 2011). So, these recent changes in Seoul Korean are consistent with a process of tonogenesis, whereby consonant-induced f0 perturbation is exaggerated and reinterpreted by learners as a primary difference, eventually replacing the original voicing or phonation difference of consonants (Hombert, Ohala, & Ewan, 1979; Thurgood, 2002; Kingston, 2011).

Several studies directly examined the change in stop VOT values in apparent time (Bailey, Wikle, Tillery, & Sand, 1991) by comparing the stop production of younger and older Seoul speakers and most found a trend of lowering VOT for aspirated stops and concomitant reduction of VOT difference between aspirated and lenis stops (Silva, 2006; Wright, 2007; Kang & Guion, 2008). Silva (2006) in particular found that of the 36 Seoul speakers examined in the study, those born before 1965 tended to maintain the VOT distinction between aspirated and lenis stops while those born after 1965 tended to merge the two categories with some younger speakers even showing a longer mean VOT value for lenis than aspirated stops. Kim (2008), on the other hand, did not replicate the age effect found in other studies. Kim (2008) examined stop production of 9 female Seoul Korean speakers ranging in age from 20s to 50s and most speakers produced a merged or reversed VOT pattern for aspirated and lenis stops, including some born before 1965. But, the speakers of the study were limited to a small number of female speakers making the results inconclusive.

It is notable, however, that studies that found an age effect on VOT realization were conducted in the US (Silva, 2006; Wright, 2007; Kang & Guion, 2008) and there are reasons to be cautious about extrapolating these results to be true of Seoul Korean as spoken by Seoul residents. Speakers in diaspora may be more conservative than speakers in the homeland due to limited exposure to sound change under way in the homeland variety or may develop traits that are distinctive from the homeland variety as it comes into contact with other languages (Nützel & Salmons, 2011). Pearce (2009) shows that speakers of Kera, a Chadic language in the process of tonogenetic sound change, use different weighting of VOT and f0 cues in their production and perception depending on their degree of contact with French. Moreover, even for speakers who are short-term visitors to a foreign country, recent research shows that adult speakers' speech continues to change as they come into contact with another language well after the critical period, in a phenomenon known as "phonetic drift" and VOT figured prominently in this line of research (Flege, 1987; Flege & Eefting, 1987; Sancier & Fowler, 1997; Kang & Guion, 2006; Fowler, Sramko, Ostry, Rowland, & Hallé, 2008; Chang, 2012). Therefore, one cannot assume that the speech pattern of Seoul residents is equivalent to that of US residents, particularly for acoustic variables like VOT, which is known to be susceptible to change due to immediate linguistic experience (Nielson, 2011; Chang, 2012). For these reasons, a systematic study of the phenomena based on data gathered from Seoul residents is essential to establish a firm empirical foundation of the phenomenon.

It is also suggested that VOT realization of stops is modulated by gender. Oh (2011) shows that young female Seoul Korean speakers show substantially shorter VOT values for aspirated stops than their male counterparts and VOT of aspirated and lenis stops overlap more in females than in males. This gender effect is unexpected given the fact that shorter VOT values are expected for males because their physiological characteristics—longer vocal folds and larger supraglottal cavity—facilitate vocal fold vibration (See references cited in Oh 2011). As Oh (2011) observes, assuming that Seoul Korean is undergoing merger of VOT in aspirated and lenis categories, the observed gender pattern is in line with the fact that sound change is commonly led by female speakers (Labov 1990). Silva (2006), on the other hand, did not find any gender effect. Kong et al. (2011) do not report any gender-based difference in VOT realization in production but found that in a perception study, listeners were sensitive to VOT cues for aspirated-lenis distinction when speakers were young males but not when the speakers were young females.

Much less is known about the development of consonant-induced f0 distinction in Seoul Korean. Studies from the 1960s already report different f0 realization at vowel onset following aspirated and lenis stops. Silva (2006) did not find any age effect on the pattern of f0 distinction in stop categories and states that "the tonal pattern has been stable over time". Oh (2011) did not find any gender effect on the realization of f0 distinction (in semitone scale) in aspirated and lenis stops and concludes that loss of VOT distinction in females is not compensated by increased difference in the f0 dimension. These studies suggest that f0 distinction has been in place for quite some time and well ahead of VOT merger. Kang and Guion (2008), on the other hand, found that in clear speech, older Korean speakers only rely on the VOT dimension to enhance aspirated-lenis contrast, while younger speakers, who tend to merge VOT of the two stop categories, use both the VOT and f0 dimensions, indicating a change in the status of f0 in the contrast. Also, in perception, Kong et al. (2011) found that listeners are far more sensitive to f0 cues in distinguishing the aspirated-lenis contrast when the speakers are female than male indicating a different status of f0 in laryngeal contrast across speakers' gender.

In other words, it remains to be seen whether the timing of tonal development and VOT merger occur in separate stages such that VOT merger only kicked in after the tonal contrast is already established and stabilized or if the tonal distinction continues to develop to compensate for the loss of VOT distinction. If the latter is the case, we expect to observe age or gender-based variation in f0 realization and f0 distinction between the aspirated and lenis stops are further strengthened concurrent with the reduction of VOT difference between the stop categories. If the increase in f0 distinction is indeed adaptive to the loss of VOT contrast, further questions arise. Is the f0 variation specific to the aspirated-lenis contrast such that increased f0 contrast is specifically targeted to compensate for the loss of VOT distinction? Or does f0 enhancement target the contrast at a more abstract level, between all high-f0 inducing contexts, including fortis stops and /h/ as well as aspirated stops ([+stiff vocal folds] in (1)) and all low-tone inducing contexts ([-stiff vocal folds], in (1))? Note that fortis stops are well separated from lenis stops in their low VOT and low H1-H2 values and therefore, they do not need the enhancement of contrast by further f0 raising. In Kang and Guion's (2008) study, where speakers are prompted to "read in a clear way, as if speaking to a foreigner audience" (p. 3915), younger speakers further increased the f0 contrast between lenis and aspirated stop conditions ([t] vs. [$t^h$]) but not the f0 contrast between [n] and [h]-initial conditions. This suggests that the f0 enhancement in clear speech specifically targets the contrast that is considered to be difficult for the listeners (Lindblom, 1990). But, what remains unknown is if the enhancement of f0 contrast as sound change in Seoul Korean, as reflected in the inter-generational difference, specifically targets the threatened contrast (lenis vs. aspirated stops) or if the enhancement targets a more abstract level of representation, i.e., distinctive features, targeting the contrast between all low tone-inducing [-stiff vocal folds] conditions and all high-tone inducing [+stiff vocal folds] conditions equally. Also, it is not clear how long after the release of stops the consonant-induced f0 perturbation extends depending on the age and gender of speakers. If the tonal pattern is still in the process of establishing itself, we expect consonant-based perturbation to be more short-lived for older or male speakers than for younger or female speakers.

In summary, while there is evidence that VOT and, to a lesser extent, f0 of laryngeal contrasts vary as conditioned by the age and gender of speakers in Seoul Korean, the exact pattern of variation needs further investigation. In the current study, we examine the production pattern of 117 speakers of Seoul Korean balanced for gender and age using a speech corpus published by the National Institute of the Korean Language (NIKL 2005). The current study employs by far the largest number of speakers and it is also the only study to date that examines the effect of both gender and age on stop realization based on data gathered from Seoul residents.

Previous studies on tonogenetic sound change tend to examine the sound pattern at a static point in time (DiCanio, 2012; Mazaudon & Michaud 2008), examine different endpoints of sound change via dialectal comparison (Svantesson & House, 2006; Brunelle, 2009a,b) or be limited in the number, gender, and age range of speakers they employ (Abramson, L-Thongkum, & Nye, 2004; Pearce, 2009; Hyslop, 2009). The current study examines sound change in progress employing a large number of speakers from a single dialect, providing a dynamic look at how the VOT merger and the tonal contrast development unfold over time.

## 2.0 Materials and methods

### *2.1 Speakers and Materials*

The data are drawn from *The Speech Corpus of Reading-Style Standard Korean* created by the National Institute of the Korean Language (NIKL 2005), which contains read speech of 60 male and 60 female speakers of the Seoul dialect residing in Seoul metropolitan area, whose parents were also speakers of Seoul dialect and residents of Seoul metropolitan area. The speech material consists of 19 well-known short stories and essays containing a total of 930 sentences. Each speaker read the texts at a comfortable speed and each sentence is stored as a separate .wav file in the corpus. The age of the speakers ranged from 19 to 71 at the time of recording in 2003. In the analysis below we convert speakers' age to their year of birth (range: 1932~ 1984) by subtracting speakers' age from 2003 to make the comparison with other studies easier. For three of the speakers, the particular sentences selected for analysis were either missing or had errors in the sound file. So, our analysis is based on the data of 117 speakers. The breakdown of the speakers in the current data by gender and age are given in **Table 1**. The numbers in parentheses are the counts for the overall corpus.

**Table 1** ***Age and gender of speakers in the NIKL Corpus***

| | 1930s | 1940s | 1950s | 1960s | 1970s | 1980s | Total |
|---|---|---|---|---|---|---|---|
| **Male** | 4 | 12 | 4 | 8 | 25 (27) | 5 | 58 |
| **Female** | 2 | 9 (10) | 25 | 3 | 11 | 9 | 59 |

Nine sentences which begin with one of nine stops of Korean are selected. The words that contain the target stop are listed in  As we use mixed effects modelling for our statistical analyses, no data pruning is necessary on account of missing data (Baayen, Davidson, & Bates, 2008).

**Table 2**. The words were chosen so that the vowel following the stop is of the same height within a given place of articulation but this was not possible for labial stops as there were very few sentences that begin with aspirated or fortis stops in general and particularly for labial stops. While this is not an ideal situation, since our main goal is to examine the effect of speakers' age and gender on the acoustic realization of stops and the contrast between them, rather than the contrast between stops per se, the current data serves our purpose well despite this shortcoming.

To examine if any f0 variation in the stop contrasts is due to system-wide change in tonal realization or variation specific to aspirated-lenis stop contrast, sentences beginning with a low-tone inducing sonorant-initial word, /nolli-ko/ "to tease-and" and a high-tone inducing /h/-initial word, /honca (nam-ɨn)/ "(left) alone", were also included. The VOT was also measured for /h/-initial words, to examine if the change in VOT of aspirated stops also affect the aspiration in /h/. A total of 1,250 tokens (11 sentences * 117 speakers – 37 missing or incorrect files) were analysed. As we use mixed effects modelling for our statistical analyses, no data pruning is necessary on account of missing data (Baayen, Davidson, & Bates, 2008).

**Table 2** ***Target words in the NIKL data (number of tokens)***

| | **Labial** | **Coronal** | **Dorsal** |
|---|---|---|---|
| **Lenis** | palk’ɨt$^h$ɨro (115)<br>"with a tip of foot" | tolaponi (116)<br>"as one looks back" | kwisini (115)<br>"ghost, NOM." |
| **Aspirated** | p$^h$iramina (107)<br>"pale club or" | t$^h$ok’inɨn (113)<br>"as for rabbit" | k$^h$ɨkocakɨn (116)<br>"of various sizes" |
| **Fortis** | p’omnætɨsi (114)<br>"as if showing off " | t’okk’at$^h$ɨn (113)<br>"same" | k’ums’okesʌ (114)<br>"in one's dream " |
| **/h/** | honca namɨn (114)<br>"left alone" | **/n/** | nolliko (116)<br>"tease and" |

*2.2 Measurements*

The sampling rate of the recording was 16,000Hz. For each target word, the onset of stop release, the onset of voicing on the following vowel as well as the offset of the vowel was manually identified. All acoustic measurements were taken in Praat (Boersma & Weenink, 2011). The onset and the offset of the second vowel of the word were also identified. The measurements included the VOT of the stop, and the duration and f0 of the first two vowels of the target word (V1 and V2). The vowel duration was measured as an indicator of speech rate and f0 of V2 was measured to examine the extent to which the stop-induced f0 variation extends into the later part of the word in different gender and age groups. VOT was defined as the duration from the onset of the stop release to the onset of the first periodic cycle of the following vowel (Ladefoged, 2003). The f0 measurements were taken at the midpoint of vowels using *Praat*'s f0 tracking function with f0 range set at 75-300 Hz for male speech and at 100-500 Hz for female speech. Acoustic measurements were taken using the script function of *Praat* followed by a manual check of all the tokens for an f0 tracking error.

F0 measurements following fortis stops often produced an octave error where the f0 track abruptly changes by an octave due to irregularity in glottal pulse. All f0 tracks were visually examined and those with an octave error were corrected to double the value *Praat* provided. 43 and 9 such corrections are made for V1 and V2 measurements, respectively. Alternative methods of f0 measurements used in the literature—taking measurements at the vowel onset, taking the mean f0 of the entire vowel duration, and taking the reciprocal of distance of two adjacent pitch pulses at onset or midpoint of vowel—were also tried and they provide comparable results as far as the effects of age and gender are concerned and are not reported separately here. For statistical analyses, f0 measurements in Hz are converted to semitones (St), a logarithmic scale of f0, to allow for comparison of f0 range across gender and age (Whalen and Levitt 1995, Oh 2011). St was calculated with 100Hz as a reference f0 using the formula, $\log_2(\text{Hz}/100)*12$.

## 3. Results

### 3.1 VOT

*3.1.1 Age and Gender effects*
This section examines the effect of age and gender on the VOT of stop categories. Statistical analyses are based on mixed-effects modelling (Pinheiro & Bates 2000; Baayen, Davidson, & Bates, 2008) using the *lmer* function in the *lme4* package (Bates, Maechler, & Bolker, 2011) for *R* (R Development Core Team, 2011). Separate linear mixed effects models are built for each of the three stop types. In these models, VOT (ms) is the response variable, YOB (year of birth) and Gender and their interactions are fixed effect predictors, and by-speaker adjustment to intercept is a random effect. YOB is centred at 1961, the mean YOB of all speakers and Gender is contrast-coded such that male is the reference category and the intercept represents the grand mean (Male: Gender= -0.5; Female: Gender=0.5). For fixed effects, *p*-values based on Markov Chain Monte Carlo (MCMC) sampling are reported. Table 3. summarizes the output of the models and Fig. 2. provides scatterplots of mean VOT of stops aggregated in 10 year bands according to speakers' year of birth and gender.

Table 3 *The Output of Linear Mixed Effects Model of VOT for three stop categories*: male is the reference category for gender and YOB is centred at 1961.

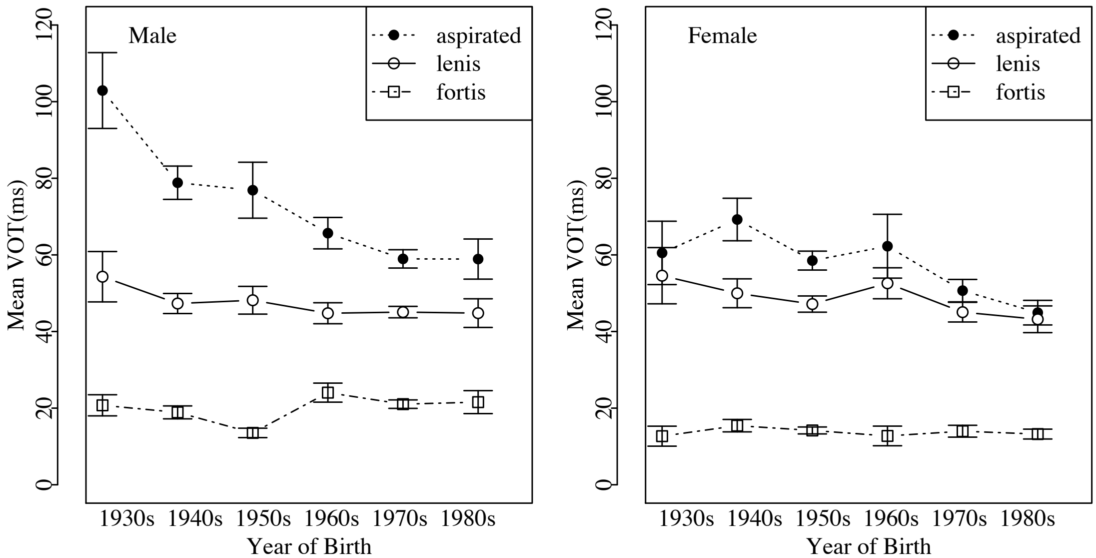
| Aspirated | Estimate | St. error | t-value | p(MCMC) |
|---|---|---|---|---|
| **(intercept)** | 63.76 | 1.21 | 52.85 | 0.0001 *** |
| **Gender** | -13.31 | 2.41 | -5.52 | 0.0001 *** |
| **YOB** | -0.65 | 0 .09 | -7.59 | 0.0001 *** |
| **Gender*YOB** | 0.27 | 0.17 | 1.60 | 0.1136 |

| Lenis | Estimate | St. error | t-value | p(MCMC) |
|---|---|---|---|---|
| **(intercept)** | 45.95 | 0.87 | 54.16 | 0.0001 *** |
| **Gender** | 0.48 | 1.73 | 0.28 | 0.7906 |
| **YOB** | -0.16 | 0.06 | -2.59 | 0.0136 * |
| **Gender*YOB** | -0.04 | 0.12 | -0.32 | 0.7414 |

| Fortis | Estimate | St. error | t-value | p(MCMC) |
|---|---|---|---|---|
| **(intercept)** | 17.18 | 0.49 | 35.34 | 0.0001 *** |
| **Gender** | -6.23 | 0.97 | -6.40 | 0.0001 *** |
| **YOB** | 0.03 | 0.03 | 0.79 | 0.4304 |
| **Gender*YOB** | -0.09 | 0.07 | -1.34 | 0.1762 |

Significance codes: ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05.

**Fig. 2.**
By-speaker mean VOT of three stop categories by speakers' year of birth, aggregated by 10 year bands, for male (left panel) and female (right panel) speakers. The error bars represent 95% confidence interval.




The aspirated stop model shows significant main effects of Gender (p=0.0001) and YOB (p=0.0001). In other words, as is observable in Fig. 2., the VOT of aspirated stops is shorter for younger than older speakers and shorter for female than male speakers. The interaction between Gender and YOB did not reach statistical significance (p=0.1136). The estimated mean VOT value is 89.27 ms (=63.76-13.31*(-0.5)-0.65*(1932-1961)) for male speakers born in 1932 and the value is reduced to 55.47 ms for male speakers born in 1984. For female speakers, the estimated mean VOT value is 13.31 ms higher than male speakers and the estimated VOT is 75.96 ms for speakers born in 1932 and 42.16 ms for speakers born in 1984.

The effects of Gender and YOB on the other two stop categories are not consistent or not as substantial as those on aspirated stops. For lenis stops, the main effect of gender is not significant. While the main effect of YOB is significant (p=0.0136)—younger speakers produce lenis stops with a shorter VOT—the size of the effect is only -0.16 ms/YOB. The effect size amounts to only 8.32 ms difference between the youngest and the oldest speakers in the study, and this is far smaller than the estimated VOT difference between the youngest and the oldest speakers for aspirated stops (33.80 ms). The interaction of YOB and Gender is also not significant. Fortis stops show a significant main effect of gender (p=0.0001)—female speakers on average produce fortis stops with a shorter VOT than male speakers and the estimated difference is 6.23 ms, which is much smaller than the estimated gender effect on the VOT values of aspirated stops (13.31 ms). The main effect of YOB is not significant and the interaction of YOB and Gender is also not significant for fortis stops.

*3.1.2 Speech rate and VOT*
As the VOT values of stops are known to vary as a function of speech rate (Kessinger & Blumstein, 1997; Oh, 2009), we need to examine the possibility that the gender and age effects on the realization of aspirated stops are merely an epiphenomenon of difference in speech rates, not a difference specific to aspirated stops. In other words, it might be the case that the older speakers in general have a slower speech rate and therefore the VOT of aspirated stops is generally longer

than that of their younger counterparts. A similar hypothesis can be entertained regarding the gender effect, i.e., women speak faster than men and therefore their VOT tends to be shorter. The fact that aspirated stops are affected significantly more than lenis or fortis stops by gender and age already provides some argument against these hypotheses but as speech rate may affect long-lag stops disproportionately (Kessinger & Blumstein, 1997), the rate-based hypothesis requires further consideration.

To test these hypotheses, the speech rate is operationalized as duration of first two vowels of the words (V1 and V2) and the effect of Gender and YOB on vowel duration was examined using mixed-effects models. These models were identical to the VOT models in section 3.1.1 except that the duration (ms) of V1 and V2 was the response variable. The output of the models is summarized in Table 4.

Table 4 *The Output of Linear Mixed Effects Model of V1 and V2:* male is the reference category for gender and YOB is centered at 1961.

| **V1** | **Estimate** | **St. error** | **t-value** | **p(MCMC)** |
|---|---|---|---|---|
| **(intercept)** | 55.20 | 0.67 | 82.45 | 0.0001*** |
| **Gender** | 2.33 | 1.34 | 1.74 | 0.0850 |
| **YOB** | -0.41 | 0.05 | -8.65 | 0.0001*** |
| **Gender*YOB** | 0.14 | 0.09 | 1.52 | 0.1320 |

| **V2** | **Estimate** | **St. error** | **t-value** | **p(MCMC)** |
|---|---|---|---|---|
| **(intercept)** | 70.86 | 0.98 | 72.86 | 0.0001*** |
| **Gender** | 7.66 | 1.95 | 3.94 | 0.0001** |
| **YOB** | -0.26 | 0.07 | -3.76 | 0.0006** |
| **Gender*YOB** | -0.01 | 0.14 | -0.04 | 0.9584 |

Significance codes: '***' 0.001 '**' 0.01 '*' 0.05.

The results show that the Gender effect on vowel duration is the opposite of what the speech rate hypothesis predicts; female speakers have longer vowels than male speakers with the effect reaching statistical significance for V2 (V1: p=0.0850; V2: p=0.0001). In other words, the female speakers produced a shorter VOT for aspirated stops than male speakers even though in general their speech rate, as measured by vowel duration, was slower than that of male speakers. So, we can be sure that the gender effect on VOT of aspirated stops is not a function of speech rate in disguise (cf. Oh 2011). As for the YOB effect, indeed older speakers produce longer vowels than younger speakers and this was true of both V1 (p=0.0001) and V2 (p=0.0006). So, the potential role of speech rate on variation in VOT of aspirated stops needs further investigation.[1]

To verify that the VOT of aspirated stops varies as a function of YOB independent of speech rate, linear mixed-effects models of aspirated stop VOT were built that were identical to the models in section 3.1.1 except that the VOT values were normalized by dividing the VOT of a stop by the duration of V1 or V2 in the same token. The YOB remained a significant predictor of VOT of

[1] It is possible that the age effect on vowel duration may be itself an indication of another sound change, i.e., loss of long vowels in younger speakers (Sohn, 1999). As the vowel length contrast is generally assumed to be only available in the first syllable of a word in Korean, we measured the duration of the second vowel as well as the first vowel. The estimated increase in vowel length per one year decrease in YOB is 0.35ms for V1 and 0.26 ms for V2. The mean vowel length of the vowels is 53.99ms (V1) and 70.02 ms (V2) and the different YOB effect size on V1 and V2 cannot be attributed to their difference in overall length. So, it seems that the age effect on the first vowel may be a combination of the effect of speech rate and a possible sound change in vowel length.

aspirated stops in both models (normalized by V1 duration: $\beta$=-0.013, SE=0.004, t=-3.351, p=0.0008; normalized by V2 duration: $\beta$=-0.005, SE=0.002, t=-3.14, p=0.0020); in both models, normalized VOT is significantly shorter for younger speakers. Not surprisingly, Gender also remained significant (normalized by V1 duration: $\beta$=-0.427, SE=0.108, t=-3.953, p=0.0001; normalized by V2 duration: $\beta$=-0.272, SE=0.044, t=-6.13, p=0.0001). The interaction of Gender and YOB was not significant in either model. In short, even when the VOT values are normalized for vowel duration, gender and age effects on VOT of aspirated stops still remain significant.

In sum, the results confirm previous observations that VOT of aspirated stops are shortening in younger Seoul Korean speakers and the YOB effect is true of both male and female speakers contra Kim (2008). The results also confirm Oh's (2011) observation that there is a substantial gender difference in VOT values, with females overall showing a shorter VOT for aspirated stops than males and this is in contrast to Silva's (2006) results which did not find any main effect of Gender.

*3.1.3 Age and Gender on VOT distinction*

To examine how the size of the VOT difference between aspirated and lenis stops varies depending on the age and gender of speakers, a linear mixed-effects model was built with YOB, Gender, Laryngeal and their interactions as fixed effects. Here, we are mainly interested in the interaction of Laryngeal factor with YOB and with Gender. By-speaker adjustment to intercept is a random effect. By-speaker random slope is determined not to be statistically significant based on a likelihood ratio test. **Table 5** summarizes the output of the VOT models along with the p-value based on MCMC sampling. Fig. 3. shows the by-speaker mean VOT difference between aspirated and lenis stops aggregated in 10 year bands of year of birth.

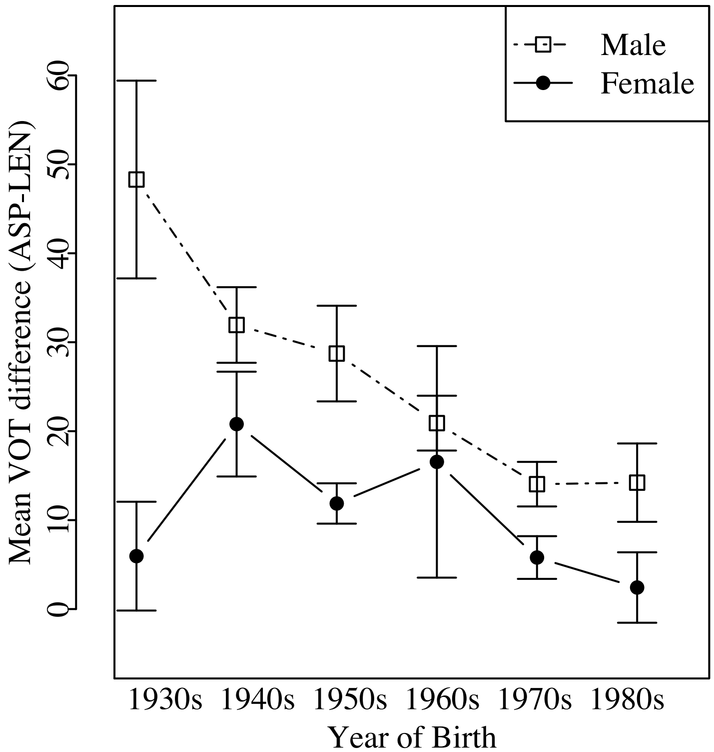
**Table 5 *The Output of Linear Mixed Effects Model of VOT*: lenis is the reference category for laryngeal factor and male is the reference category for gender.**

| | **Estimate** | **St. error** | **t-value** | **p(MCMC)** |
|---|---|---|---|---|
| **(intercept)** | 55.35 | 0.86 | 64.31 | 0.0001 *** |
| **Gender** | -6.40 | 1.72 | -3.72 | 0.0001 *** |
| **YOB** | -0.41 | 0.06 | -6.66 | 0.0001 *** |
| **Laryngeal** | 16.83 | 1.42 | 11.82 | 0.0001 *** |
| **Gender*YOB** | 0.11 | 0.12 | 0.92 | 0.2894 |
| **Gender*Laryngeal** | -13.76 | 2.85 | -4.83 | 0.0001 *** |
| **YOB*Laryngeal** | -0.49 | 0.10 | -4.88 | 0.0001 *** |
| **Gender*YOB*Laryngeal** | 0.30 | 0.20 | 1.50 | 0.1326 |

Significance codes: ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05.

**Fig. 3.**

By-speaker difference in mean VOT between aspirated and lenis stops plotted against speakers' YOB and gender.




The main effect of Laryngeal is significant (p=0.0001) indicating that on average the VOT value of aspirated stops is longer than that of lenis stops but this contrast is modulated by both Gender and YOB. The significant interaction of Gender*Laryngeal (p=0.0001) indicates that the VOT difference between lenis and aspirated stops is reduced for females compared to males by 13.76 ms; the difference is 23.71ms (=16.83-13.76*(-0.5)) for males but only 9.95 ms (=16.83-13.76*0.5) for females. This pattern is visually represented in Fig. 3. where the mean VOT differences for male speakers (a dotted line) are higher than those for female speakers (a solid line). In fact, seven of the 59 female speakers show higher mean VOT values for lenis than aspirated stops including four speakers born before 1965. For male speakers, on the other hand, only one speaker born in 1977 showed this pattern. The significant interaction of YOB*Laryngeal (p=0.0001) shows that the contrast is reduced for younger speakers compared to older speakers. So, for speakers born in 1932, the aspirated and lenis stops are estimated to differ in VOT by 31.04ms (=16.83-0.49*(1932-1961)) but the estimated difference is reduced to only 5.56ms (=16.83-0.49*(1984-1961)) for speakers born in 1984. This pattern is visible in Fig. 3**.**, where the mean VOT difference decreases in younger speakers. The three-way interaction of Gender*YOB*Laryngeal is not significant. To summarize, the results confirm previous reports that younger speakers reduce the VOT difference between aspirated and lenis stops and also confirm that the female speakers in general are ahead of the males in this trend.

*3.1.4 Aspirated stops and /h/*

The duration of aspiration in /h/ was also measured to examine if /h/ patterns with the aspirated stops in VOT variation. As shown in Fig. 4, change in the VOT of /h/ closely follows that of aspirated stops. A linear mixed-effects model was built with VOT as a dependent variable and YOB, Gender, consonant type (/h/ vs. aspirated stops), and their interactions as fixed effects. By-speaker adjustment to the intercept was the only significant random effect. The result is summarized in **Table 6**. There is no significant interaction of consonant type with YOB or

Gender, indicating that the VOT values of /h/ and aspirated stops are moving parallel to each other.

**Fig. 4.**
By-speaker mean VOT of three stop categories by speakers' year of birth, aggregated by 10 year bands, for male (left panel) and female (right panel) speakers. The error bars represent 95% confidence interval.

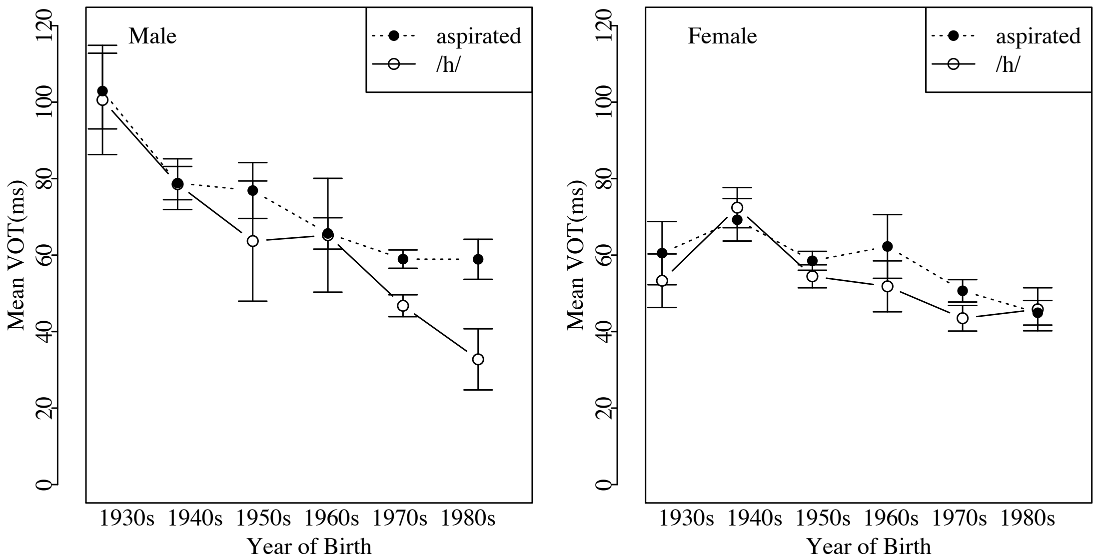



**Table 6** ***The Output of Linear Mixed Effects Model of VOT*****: aspirated stops is the reference category for consonant type (/h/ is the other category) and male is the reference category for gender.**

| | Estimate | St. error | t-value | p(MCMC) |
|---|---|---|---|---|
| **(intercept)** | 66.64 | 1.66 | 40.09 | 0.0001 *** |
| **Gender** | -11.20 | 2.32 | -4.82 | 0.0001 *** |
| **YOB** | -0.98 | 0.11 | -9.02 | 0.0001 *** |
| **ConsonantType** | -7.56 | 3.32 | -2.27 | 0.0235 * |
| **Gender*YOB** | 0.44 | 0.16 | 2.71 | 0.0071 ** |
| **Gender*ConsonantType** | 4.22 | 4.65 | 0.91 | 0.3640 |
| **YOB* ConsonantType** | -0.39 | 0.22 | -1.80 | 0.0723 |
| **Gender*YOB* ConsonantType** | 0.33 | 0.33 | 1.03 | 0.3056 |

Significance codes: '***' 0.001 '**' 0.01 '*' 0.05.

**3.2 F0**

*3.2.1 Gender and age effects on f0 of V1*
In this section, we examine how the consonant-induced f0 distinction on the vowel immediately following the consonant (V1) varies as a function of age and gender. Fig. 5. shows the by-speaker mean f0 values for each stop category aggregated in 10 year bands of speakers' year of birth. The mean f0 values following initial /h/ and /n/ are also measured to examine if the age- and gender-based f0 variation, if any, selectively targets the lenis-aspirated contrast or the effect affects a larger class of sounds. **Fig. 6** shows the by-speaker difference in mean f0 values between lenis-aspirated, lenis-fortis, and /n/-/h/ pairs, aggregated in 10 year bands.

**Fig. 5.**
Mean f0 (St) at V1 midpoint by Year of Birth and Gender.

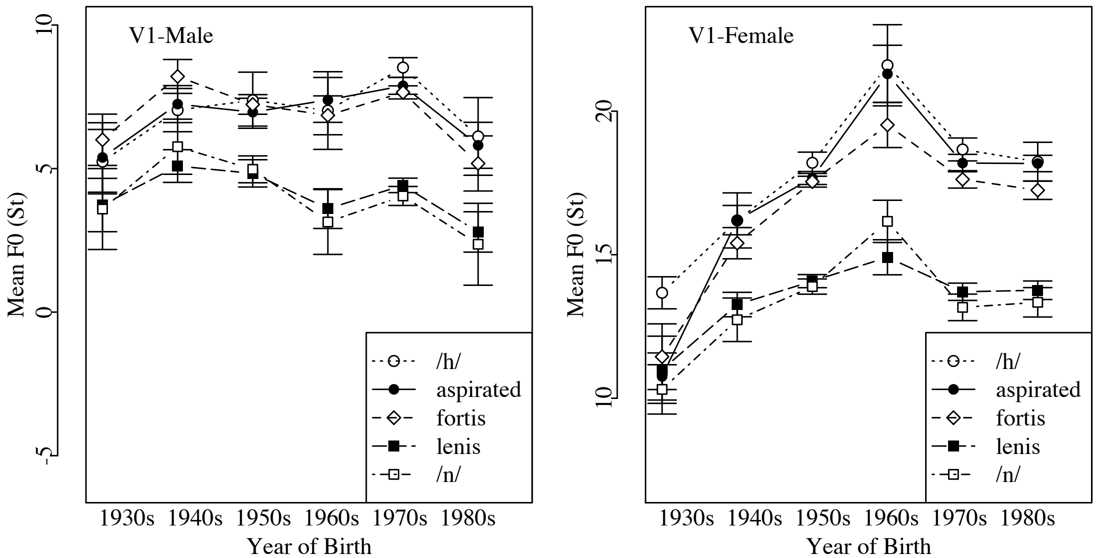
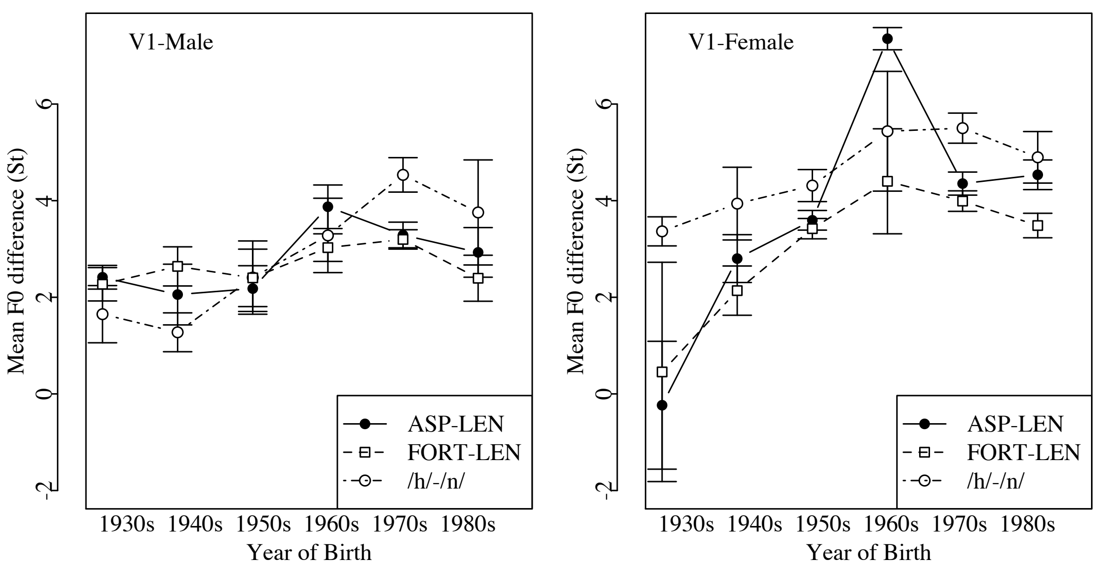



**Fig. 6**
Mean difference in f0 (St) between L and H contexts at V1 midpoint by Year of Birth and Gender.




To examine how the size of the f0 difference between stop categories (aspirated vs. lenis and fortis vs. lenis) varies depending on the age and gender of speakers, linear mixed-effects models were built with YOB, Gender, Laryngeal, and their interactions as fixed effects. By-subject adjustment to the intercept and the slope for Laryngeal were both included in the model. As MCMC-based p-value estimation is not available in the *lmer* function for this more complex model, we used the t-statistics with a degree of freedom of 1000 for to determine the p-values for this model (Baayen, et al., 2008). The Laryngeal factor has three levels and they were coded so that the lenis category is the reference for comparison and the intercept represents the grand mean (lenis: LaryngealAsp=-1/3, LaryngealFort=-1/3; aspirated: LaryngealAsp=2/3, LaryngealFort= -1/3; fortis: LaryngealAsp=-1/3, LaryngealFort=2/3). **Table 7** summarizes the output of the VOT and f0 models along with the p-values.

**Table 7** ***The Output of Linear Mixed Effects Models of f0*****: lenis is the reference category for laryngeal factor and male is the reference category for gender.**

| | Estimate | St. error | t-value | Pr(>\|t\|) |
|---|---|---|---|---|
| **(intercept)** | 11.22 | 0.21 | 53.89 | 0.0001 *** |
| **Gender** | 9.78 | 0.42 | 23.48 | 0.0001 *** |
| **YOB** | 0.02 | 0.01 | 1.51 | 0.1276 |
| **LaryngealAsp** | 3.33 | 0.13 | 26.07 | 0.0001 *** |
| **LaryngealFort** | 3.12 | 0.12 | 26.01 | 0.0001 *** |
| **Gender*YOB** | 0.05 | 0.03 | 1.66 | 0.1006 |
| **Gender*LaryngealAsp** | 0.86 | 0.26 | 3.38 | 0.0014 ** |
| **Gender*LaryngealFort** | 0.34 | 0.24 | 1.43 | 0.1435 |
| **YOB *LaryngealAsp** | 0.05 | 0.01 | 5.17 | 0.0001 *** |
| **YOB *LaryngealFort** | 0.03 | 0.01 | 3.20 | 0.0024 ** |
| **YOB*Gender*LaryngealAsp** | 0.02 | 0.02 | 1.09 | 0.2201 |
| **YOB*Gender*LaryngealFort** | 0.03 | 0.02 | 1.65 | 0.1023 |

Significance codes: '***' 0.001 '**' 0.01 '*' 0.05.

The model shows significant main effects of Laryngeal (LaryngealAsp: p=0.0001; LaryngealFort: p=0.0001) indicating that on average the f0 value of aspirated and fortis stops is higher than that of lenis stops. But, the f0 difference is modulated by both Gender and YOB. The significant interaction of Gender*LaryngealAsp (p=0.0001) indicates that the aspirated-lenis contrast in f0 is larger for females compared to males; the difference is 2.90 St (=3.33+0.86*(-0.5)) for males and 3.76 St (=3.37+0.86*0.5) for females. The significant interaction of YOB*LaryngealAsp (p=0.0001) indicates that the f0 distinction between the aspirated and lenis stop contexts increases in younger speakers, as shown by the increased gap between the mean f0 values of aspirated and lenis stop contexts in Fig. 5. and by the upward trend of the mean f0 difference between aspirated and lenis contexts in **Fig. 6**. For speakers born in 1932, the aspirated and lenis stops are estimated to differ by 1.88 St (=3.33+0.05*(1932-1961)) and the estimated difference is increased to 4.48 St (=3.37+0.05*(1984-1961)) for speakers born in 1984.

For the lenis-fortis contrast, the significant interaction of YOB*LaryngealFort (p=0.0024) indicates that f0 contrast is larger for younger than older speakers. This suggests that the enhancement of f0 distinction in younger speakers is not limited to the aspirated-lenis contrast, where VOT distinction is undergoing a merger, but also found in L vs. H contrast in general. This pattern is also visible in Fig. 5. where all H tone inducing contexts (aspirated, fortis, and /h/) and all L tone inducing contexts (lenis and /n/), respectively, move in tandem. Also, in **Fig. 6**, the f0 differences between all three L vs. H pairs are generally increasing in younger speakers. At the same time, it is notable that the f0 difference is estimated to increase more steeply in the aspirated-lenis contrast at 0.05St/YOB than in the fortis-lenis contrast at 0.03St/YOB indicating that the f0 change may not affect all H tone or L tone contexts to the same degree. The two-way interaction of Gender*LaryngealFort is not significant, indicating that unlike aspirated-lenis contrast, the fortis-lenis contrast is not further enhanced in female speakers than male speakers. The three-way interaction of Gender*YOB*LaryngealFort is not significant.

To summarize, the younger speakers in general make a larger f0 distinction between stop types compared to older speakers. This is true for the aspirated-lenis contrast and also for the lenis-fortis contrast, but to a lesser extent for the latter than for the former. Gender effect is not consistent across stop contrasts; for the aspirated-lenis contrast, females make larger f0 distinction than male speakers and this is in line with the observation that females are ahead of males in the VOT merger process. The St distinction in the fortis-lenis contrast, on the other hand, does not differ significantly by gender.

To further examine if the increase in f0 difference between aspirated and lenis contexts—where VOT merger directly threaten the contrast—are significantly different from those of other H or L tone-inducing contexts, a mixed effects model was built comparing the effect of Age and Gender on the f0 of aspirated, fortis and /h/ contexts. If the aspirated-lenis contrast is selectively targeted for f0 enhancement, we expect the f0 of aspirated stops to be affected by Age and Gender more than other H tone inducing contexts (i.e. /h/ or fortis stops). The output of the model is summarized in **Table 8**.

**Table 8** ***The Output of Linear Mixed Effects Models of f0 (Aspirated, Fortis, and /h/)*****: aspirated is the reference category for Consonant and male is the reference category for Gender.**

| | Estimate | St. error | t-value | p(MCMC) |
|---|---|---|---|---|
| **(intercept)** | 12.43 | 0.22 | 56.44 | 0.0001 *** |
| **Gender** | 10.22 | 0.44 | 23.19 | 0.0001 *** |
| **YOB** | 0.04 | 0.02 | 2.47 | 0.0001 *** |
| **Consonant/h/** | 0.29 | 0.13 | 2.18 | 0.0766 |
| **ConsonantFort** | -0.22 | 0.10 | -2.26 | 0.0966 |
| **Gender*YOB** | 0.04 | 0.03 | 1.34 | 0.0230 * |
| **Gender*Consonant/h/** | 0.42 | 0.27 | 1.58 | 0.2102 |
| **Gender*ConsonantFort** | -0.53 | 0.19 | -2.79 | 0.0304 * |
| **YOB *Consonant/h/** | 0.00 | 0.01 | 0.31 | 0.8880 |
| **YOB *ConsonantFort** | -0.02 | 0.01 | -2.94 | 0.0152 * |
| **YOB*Gender*Consonant/h/** | -0.04 | 0.02 | -2.23 | 0.1008 |
| **YOB*Gender*ConsonantFort** | 0.01 | 0.01 | 0.74 | 0.5584 |

Significance codes: '***' 0.001 '**' 0.01 '*' 0.05.

The comparison of the aspirated context with the fortis and /h/ contexts in **Table 8** shows that the main effect of Consonant type (both Consonant/h/ and ConsonantFort) is not significant, indicating that on average the f0 values in aspirated contexts do not differ significantly from those in fortis or /h/ contexts. The main effect of Gender is significant, as expected; f0 is overall higher for female than male speakers. The main effect of YOB is also significant indicating that f0 of the aspirated, fortis, and /h/ is higher for younger than older speakers. There is no significant interaction of Consonant/h/ with YOB or with Gender, which indicates that f0 values in /h/ context pattern similarly to those in the aspirated context in terms age- or gender-based variation, despite the fact that raising of f0 in /h/ context does not enhance any minimal contrast at the risk of merger.[2] ConsonantFort, on the other hand, interacts significantly with YOB (p=0.0304) and Gender (p=0.0152), indicating that f0 in fortis contexts does not increase in female or younger speakers' speech as much as in the aspirated contexts. In other words, the f0 distinction enhancement applies to all H tone-inducing contexts as a natural class but more so for /h/ and aspirated stop contexts than for fortis contexts, indicating that /h/ and aspirated contexts form a natural class of aspirated consonants apart from fortis consonants.

[2] From Section 3.1.4., we saw that the VOT values of /h/ mirror the lowering trend of VOT in aspirated stops but Even with the shortening, VOT of /h/ is long enough to distinguish /h/-initial words from vowel initial words.

*3.2.2 Gender and Age effects on f0 of V2*

Now we turn to the question of how the laryngeal feature of the initial consonant affects the realizaiton of f0 on the second syllable of the word (V2). Recall from **Error! Reference source not found.** that V2 is realized with a H tone of LH or HH boundary tones. Previous studies report that the high tone in this position is realized at an even higher f0 when the initial consonant of AP is one of the high tone-inducing categories (Silva 2006). In other words, the effect of word-initial consonant on f0 realization extends far beyond the immediately following vowel. In this section, we examine how the effect of word-initial consonant on the f0 of V2 varies as a function of age and gender. Fig. 7. shows the mean f0 values at the V2 midpoint for each consonant category aggregated in 10 year bands of speakers' year of birth. Also, a mixed-effects model was built to examine how the initial consonant affects the f0 realization of V2. The response variable is the f0 (St) at V2 midpoint and Consonant type, Gender, YOB, and their interactions are the fixed effects. For this model, we coded the consonant as either H-tone inducing (i.e. aspirated, fortis, and /h/) or L-tone inducing (i.e., lenis and /n/). **Table 9** summarizes the output.

Fig. 7.

Mean f0 at V2 midpoint for different word-initial consonant context by Year of Birth and Gender.

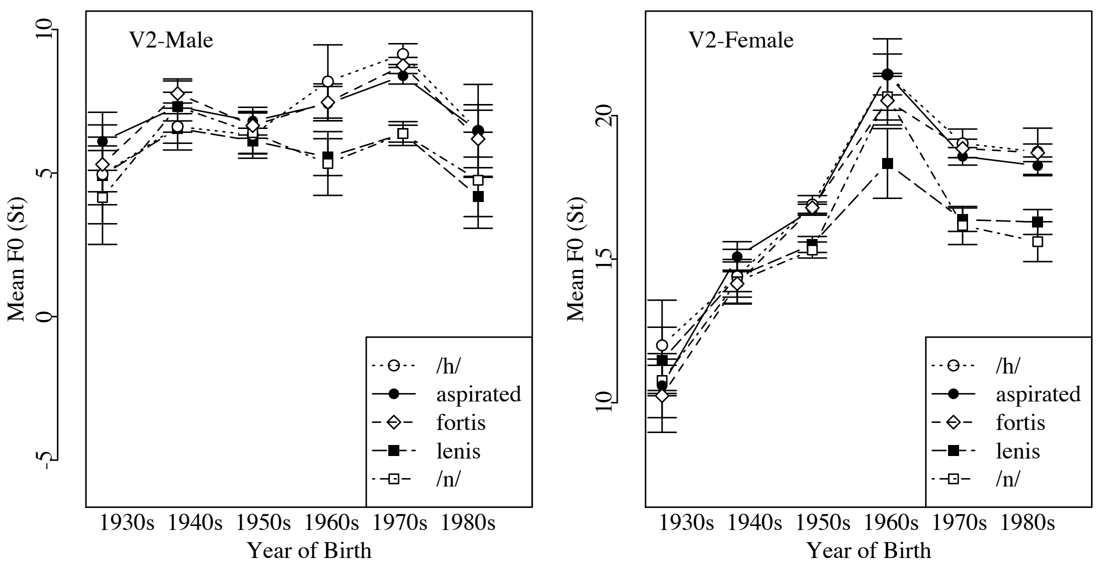



**Table 9** ***The Output of Linear Mixed Effects Models of f0 at V2:*** **L tone inducing consonants is the reference category for Consonant factor and male is the reference category for gender.**

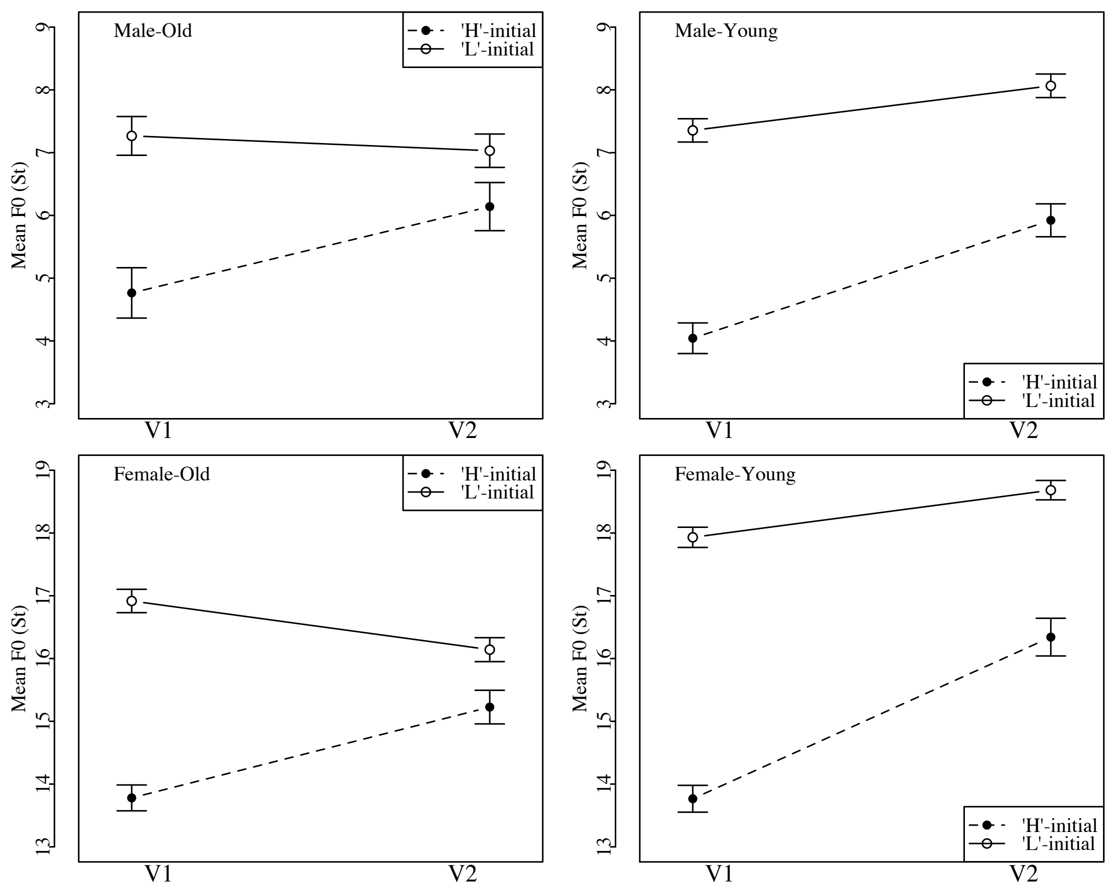
| | Estimate | St. error | t-value | p(MCMC) |
|---|---|---|---|---|
| **(intercept)** | 11.58 | 0.21 | 54.25 | 0.0001 *** |
| **Gender** | 9.43 | 0.43 | 22.08 | 0.0001 *** |
| **YOB** | 0.06 | 0.02 | 3.74 | 0.0002 *** |
| **Consonant** | 1.49 | 0.08 | 18.84 | 0.0001 *** |
| **Gender*YOB** | 0.07 | 0.03 | 2.20 | 0.0001 *** |
| **Gender*Consonant** | 0.01 | 0.16 | 0.04 | 0.9698 |
| **YOB *Consonant** | 0.05 | 0.01 | 9.77 | 0.0001 *** |
| **YOB*Gender*Consonant** | 0.01 | 0.01 | 1.26 | 0.2085 |

Significance codes: ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05.

The significant main effect of Consonant (p=0.0001) indicates that on average even in V2, the f0 values are significantly higher when the phrase-initial consonant is a H tone inducing type (i.e., HH context) than a L tone inducing type (i.e., LH context). The significant interaction of Consonant with YOB (p=0.0001) indicates that the Consonant effect is stronger in younger than older speakers; for speakers born in 1984, the estimated mean f0 values of V2 are 2.59 St (=1.49+0.05*(1984-1962)) higher in HH tone contexts than in LH tone contexts but for speakers born in 1932, the estimated difference falls to around zero (-0.01=1.49+0.05*(1932-1962)) indicating that the effect of phrase-initial consonants on f0 does not reach far enough to affect V2 for older speakers.

Fig. 8 schematically represents the f0 trajectory in words beginning with H tone inducing consonants and L tone inducing consonants based on mean F0 values at vowel midpoints. For older speakers (left panels), the initial consonant-induced f0 difference substantially reduced in the second vowel but for younger speakers (right panels), the high tone on V2 is further raised when the initial consonant is a H tone inducing type.

**Fig. 8 The mean F0 at vowel midpoint by the word-initial consonant type for male (top row) and female (bottom row) speakers born in 1961 or before (left column) and after 1961 (right column). Error bars represent 95% confidence interval.**




Unlike in V1, where females produce a larger consonant-induced f0 contrast than males, in V2 no significant interaction of Gender*Consonant is found. From Fig. 8, however, we can see that the lack of gender difference in f0 contrast in V2 is not because younger female speakers are not raising the f0 on the V2 in HH context but because they raise f0 not only in HH context but also in LH context compared to older speakers. Recall once again that V2 carries a high tone even when the initial consonant of AP is a L tone inducing type. In other words, young female speakers are not only increasing the paradigmatic f0 contrast between LH vs. HH conditions but they are also increasing the syntagmatic contrast between the initial L tone and the following H tone in the AP-initial LH tone sequence, as indicated by the steeper slope of the solid line in the lower right panel.

*3.2.3 Speech rate and f0 contrast*

As with the age effect on VOT, we need to examine the possibility that the age effect on f0 contrast might be an epiphenomenon of a speech rate effect. As older speakers' speech rate is in general slower, the midpoint of V1 and V2 tend to occur farther from the initial consonant in time in older speakers' speech and the consonant-induced f0 contrast is expected to be weaker for older speakers.[3] Further tests are conducted to examine if the consonant-induced f0 contrast differs by age even after the speech rate is taken into account. For the model of f0 (St) at V1, a mixed-effects model is built where the fixed effects include Gender, YOB, V1 duration, Consonant Type (L tone vs. H tone) and the interaction of each of the first three with Consonant Type. The YOB is residualized against V1 duration (YOB.res) to test if the YOB effect on f0 distinction between L and H contexts is significant above and beyond any effect attributable to V1 duration. The model of f0 (St) at V2 was the same except V2 duration is used instead of V1. In both models, the interaction of YOB.res*Consonant remains significant (V1 model: $\beta=0.0211$, S.E.=0.0084, $t=2.52$, $p=0.0168$; V2 model:, $\beta=0.0525$, S.E.=0.0071, $t=7.41$, $p=0.0001$). In other words, even after the effects of vowel duration are taken into account, the effects of age on consonant-induced f0 distinction on V1 and V2 remain unchanged and younger speakers show a larger f0 difference between L and H contexts than older speakers.

**3.3 Cue weighting**

So far we examined the effect of age and gender on the two acoustic variables, VOT and f0, separately. In this section, we examine how the weighting of the two cues in signalling aspirated-lenis contrast change depending on the age and gender of speakers. Separate mixed-effects logistic regression models are built for male and female speakers. The binary response variables were the laryngeal categories and the fixed effects predictors were VOT, f0 at V1 midpoint, YOB and the interaction of each acoustic variable with YOB. The acoustic variables were normalized using z-score transformation using the *scale* function so that the effect size of the two variables can be directly compared (Kong et al., 2011). The by-speaker random intercept was added to both models and by-speaker adjustment to f0 slope was also added to the female model as it was determined to be significant by a likelihood ratio test. ($\chi^2=7.8304$, $p=0.0051$, df=1). Figure 8. provides partial effects of each acoustic variable. The absolute value of the coefficient for each acoustic variable in the model determines the steepness of the curves in the figure.

For male speakers, both VOT and f0 are significant and their coefficients are comparable in their normalized values ($\beta_{VOT}=1.407$, $p<0.0001$; $\beta_{f0}=1.197$, $p<0.0001$) indicating that male speakers use both VOT and f0 cues to distinguish aspirated from lenis stops. The significant interaction of

[3] Also, Cho (2011) shows that the anchoring position of AP boundary tones, particularly the second tone of AP initial boundary tones is affected by speech rate in Seoul Korean. Specifically, we expect the H tone to appear later in the second syllable in fast speech compared to slow speech. As our data contains both 3- and 4- syllable words, it is not apparent how the speech rate affects our observation. In the most ungenerous interpretation, we may expect that the measurements from younger (=faster) speakers overestimate the consonant effect compared to those from older speakers as the measurements are taken at a transition from the first L/H to the second H tone.

f0*YOB (p=0.0004) indicates that younger speakers weigh f0 more than older speakers. In the model for female speakers, on the other hand, VOT is not a significant factor and the interaction of VOT*YOB is not significant either indicating that overall, VOT is not a significant cue for lenis-aspirated stop contrast for female speakers regardless of speakers' age. F0 is significant (p<0.0001) and the interaction of f0 with YOB is also significant (p=0.0051). In other words, female speakers rely solely on f0 to distinguish aspirated and lenis stops and the distinction by f0 is sharper for younger than older speakers, as indicated by the steeper slope of the curve for younger speakers in Fig 8 (Female-f0).

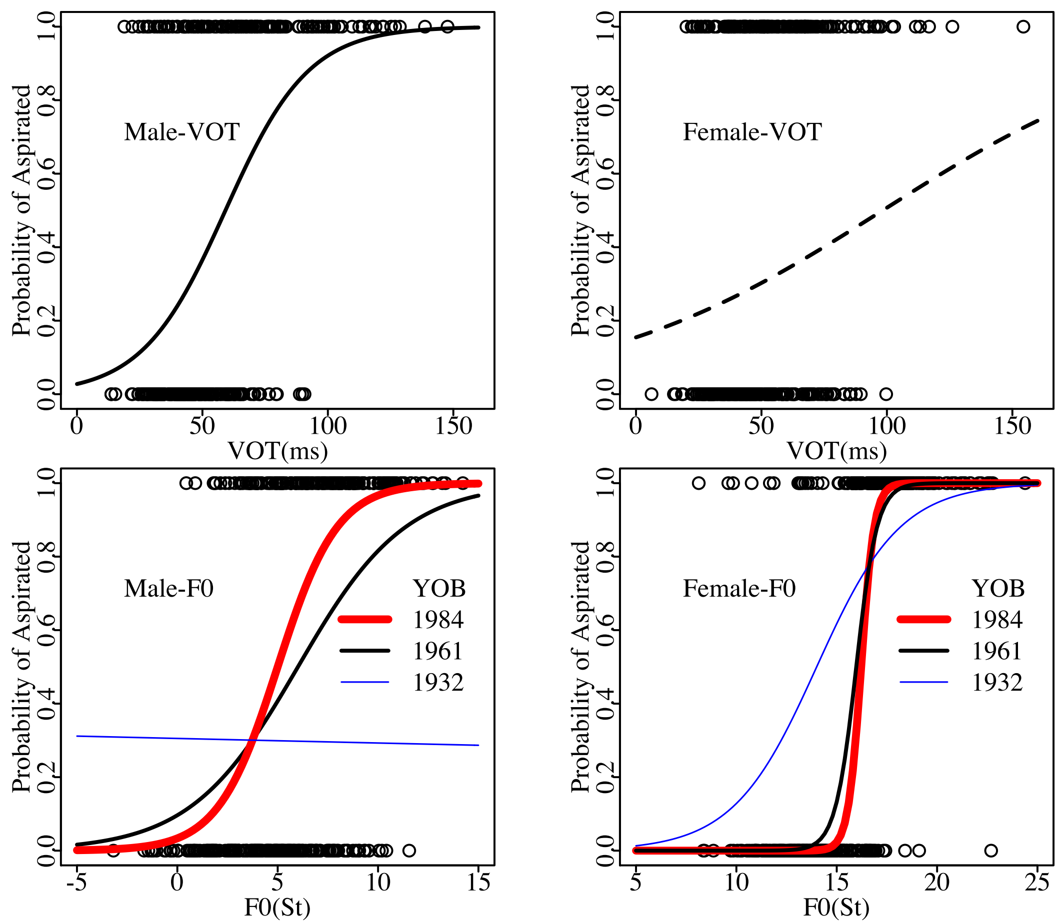
**Table 10** ***The Output of Logistic Mixed Effects Models of Aspirated-Lenis contrast:*** **lenis is the reference category for laryngeal factor and VOT and F0 are normalized.**

| ***Male*** | **Estimate** | **St. error** | **z-value** | **Pr(>\|z\|)** |
|---|---|---|---|---|
| **(intercept)** | -0.119 | 0.1591 | -0.749 | 0.4538 |
| **VOT** | 1.407 | 0.2102 | 6.691 | <0.0001*** |
| **F0** | 1.197 | 0.1880 | 6.370 | <0.0001*** |
| **YOB** | 0.025 | 0.0105 | 2.423 | 0.0154* |
| **VOT*YOB** | 0.003 | 0.0131 | 0.216 | 0.8291 |
| **F0*YOB** | 0.042 | 0.0118 | 3.541 | 0.0004*** |

| ***Female*** | **Estimate** | **St. error** | **z-value** | **Pr(>\|z\|)** |
|---|---|---|---|---|
| **(intercept)** | -0.802 | 0.401 | -2.000 | 0.0456 * |
| **VOT** | 0.352 | 0.331 | 1065 | 0.2869 |
| **F0** | 5.676 | 0.725 | 7.826 | <0.0001*** |
| **YOB** | -0.053 | 0.031 | -1.701 | 0.0889 |
| **VOT*YOB** | -0.001 | 0.025 | -0.027 | 0.9785 |
| **F0*YOB** | 0.147 | 0.052 | 2.804 | 0.0051** |

Significance codes: '***' 0.001 '**' 0.01 '*' 0.05.

**Fig. 9.**
Scatterplot of lenis (0) and aspirated (1) stops with respect to VOT and F0 values. The curves represent the probability of aspirated stops (1) estimated by the logistic regression models in **Table 10**.




**4.0 Summary and Discussion**

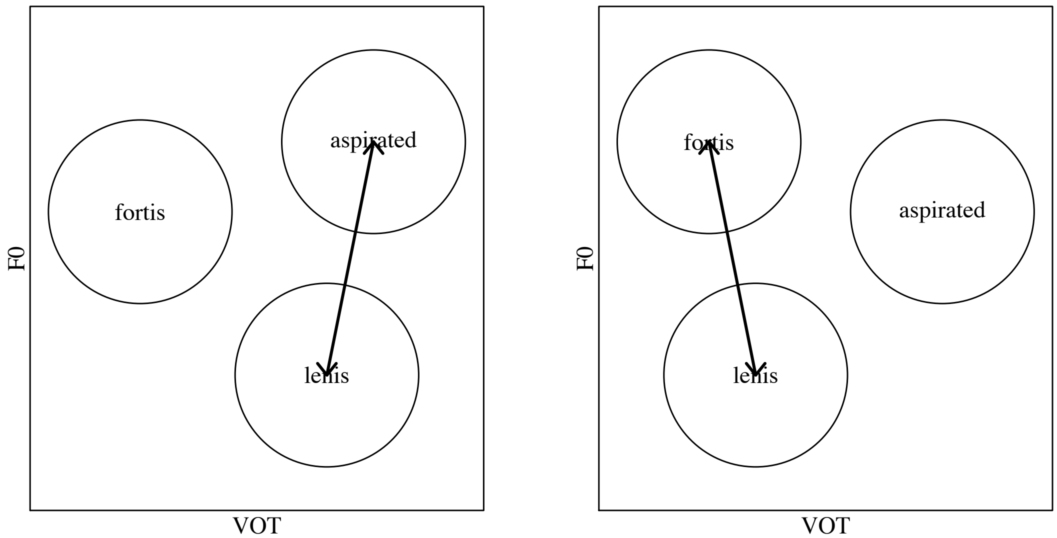
The findings of the study confirm previous observations about age- and gender-based variation in VOT values of Seoul Korean stops; VOT of aspirated stops is shorter for younger than older speakers and for female than male speakers and the VOT difference between aspirated and lenis stops is reduced in younger and female speakers accordingly. The study also presents novel findings about the effect of age and gender on consonant-induced f0 variation. While the finding on f0 largely confirms Silva (2006)'s observation that the effect of lenis vs. aspirated stop contrast on the f0 of V1 is fairly well established in both older and younger speakers' speech, a clear trend is identified whereby the f0 distinction between aspirated and lenis stops is amplified and extends further into the phrase in the speech of younger compared to older speakers and in the speech of female compared to male speakers. The trend of f0 distinction enhancement is also found for fortis-lenis contrast but to a lesser extent than for aspirated-lenis contrast. When VOT and f0 are examined together, we found that speakers use different cue weighting for aspirated-lenis stop contrast depending on their age and gender; The older males exhibit the most conservative pattern relying mainly on VOT to signal the contrast and younger males exhibit an intermediate pattern whereby VOT and f0 are both used contrastively. Finally, females, particularly the younger females, exhibit the most advanced stage of change whereby the contrast is signalled solely by the f0 difference.

Our result indicates that the development of tonal distinction and loss of VOT distinction does not necessarily occur in clear stages, whereby redundant tonal patterns get reinterpreted as primary contrast and become phonologized first, followed by a stage where the redundant VOT difference gets lost (cf. Hombert 1978), but that the two processes may take place concurrently. We also find that the f0 enhancement operates at two different levels. First of all, the f0 enhancement applies generally and the contrast between all H tone inducing contexts (aspirated, fortis, /h/) and all L tone inducing contexts (lenis, sonorants) is amplified in younger speakers' speech. At the same time, the degree of enhancement is not equal across segmental contexts; aspirated stops and /h/ pattern together in showing more amplification of f0 distinction from L tone contexts than fortis stops do. This suggests that the aspirated stops and /h/ form a natural class where the tonal distinction is further enhanced above and beyond the enhancement applied to L vs. H contrast in general. The added enhancement effect for aspirated categories suggests that as the VOT difference between aspirated and lenis stops reduces, speakers compensate for the reduced contrast by enhancing the tonal distinction further. But, it is notable that the added enhancement targets the natural class of aspirated consonants including both the aspirated stops and /h/ rather than targeting the aspirated stops only where the contrast is in danger of being merged without f0 distinction. This suggests that the tonal contrast enhancement as sound change is mediated by structural categories (i.e., all [+spread glottis] sounds) in contrast to on-line enhancement in clear speech, which narrowly targets threatened segmental contrast (Kang and Guion 2008).

It is interesting to examine the development of tonal enhancement in Seoul Korean in view of the tonal development in other Asian languages, where (partial) merger of a three-way laryngeal contrast developed into a two-way tonal split in some languages and a three-way split of tones in other languages (Haudricourt, 1972; Kingston, 2011). For a two-way split, a common pattern is for voiced consonants to induce lower tones and glottalized/unaspirated and aspirated voiceless consonants to induce higher tones, as in Tho spoken in North Vietnam (aspirated: **hma*>*ma*[H] 'dog'; unaspirated: **pi* >*pi*[H] 'year'; voiced: **ma*>*ma*[L] 'come', **bi*>*pi*[L] 'fat'). In a three-way split, often the voiced category gives rise to lower tones and aspirated and constricted categories give rise to higher tones. But, languages differ in terms of which of the two voiceless categories occupy a higher position in the tonal range. The choice seems to depend in part on how the laryngeal contrast of the original contrast merged; i.e., the unmerged category occupy the middle range and the merged categories split to occupy the higher and the lower tonal ranges. In Nakhorn Sithammarat Thai, for example, the voiced and the aspirated categories merged and the original aspirated categories developed higher tones (voiced > L, aspirated > H) while the unaspirated/glottalized category remained distinct from the merged category and not affected by the tonal enhancement (unaspirated > M), resulting in a three-way tonal split (aspirated: *hmaa*>*ma* [H] 'dog' ; unaspirated: *kin*> *kin* [M] 'to eat'; voiced: *maa*> *maa* [L] 'to come'). In Yung-chiang Kam, on the other hand, the unaspirated/glottalized consonants merged with the voiced consonants and came to occupy a higher tonal range than the aspirated consonants (voiced > L, aspirated > M, unaspirated/glottalized > H). In other words, the L vs. H contrast is reserved for the categories that underwent VOT merger while the unmerged category takes on the intermediate tone, M.

In this context, it is notable that many recent studies on Korean stops find higher f0 values in aspirated than fortis contexts (Kim, 1994; Choi, 2002; Silva, 2006; Lee & Jongman, in press). In contrast, a study of the dialects of Korean spoken by ethnic Koreans in China show that the VOT values of lenis categories are closer to that of fortis stops than aspirated stops and the f0 values for fortis stops are in general higher than those of aspirated stops (Y. Kang & Han, 2012). This dialectal difference is schematically represented in Fig. 10. The f0 distinction is further enhanced for the categories that are threatened by VOT overlap and the third category that are well separated from the others along VOT take on an intermediate f0 position.

**Fig. 10** VOT merger and compensatory enhancement of f0 contrast in three-way




Another example of trading between VOT and f0 among three-way laryngeal contrast in progress is found in Kurtöp, a Tibeto-Berman language of Bhutan (Hyslop, 2009). Kurtöp has a three-way laryngeal contrast of voiced, voiceless unaspirated, and aspirated obstruents. The tone is contrastive following sonorant-initial syllables but predictable following obstruents, with low f0 following voiced obstruents and high f0 following voiceless obstruents, both aspirated and unaspirated. Hyslop (2009) examined the production of various stop-initial syllables by one younger and one older male speaker, a generation apart, and found that the older speaker made a two-way tonal distinction with the two tonal categories differing by approximately 12 Hz. The younger speaker, on the other hand, produced a larger tonal contrast (approximately 20 Hz) and also produced a three-way split in f0 patterns with the unaspirated stops with a significantly higher f0 than the aspirated stops (unaspirated > aspirated >> voiced). At the same time, the younger speaker devoices the voiced stops more often than older speakers partially merging the contrast between voiced and unaspirated stops. In other words, here again, we see a trading of VOT and f0, where loss of VOT distinction is compensated by further exaggerated f0 contrast and this is very similar to what we observe in Korean as well. More generally, our data suggests that Seoul Korean is showing a mixture of two-way tonal split (aspirated, fortis > lenis and sonorants) and three-way tonal split (aspirated>fortis >> lenis and sonorants) patterns. If the fortis category remains unmerged from the other two categories, with respect to VOT and H1-H2 and the divergent trend of f0 enhancement between fortis and aspirated obstruents is sustained, we expect that f0 divergence will further amplify to lead to three distinct tonal levels (lenis and sonorants > L, fortis > M, aspirated > H).

To conclude, the study examined current sound change in Seoul Korean based on corpus data. In addition to confirming previous findings of change in VOT—namely, the loss of VOT difference between lenis and aspirated stops—we also uncovered the trend of f0 enhancement concurrent with the reduction of VOT distinction. Also, we also found evidence that f0 enhancement is adaptive—i.e., it preferentially targets aspirated-lenis contrast where the contrast is threatened by the VOT merger—but the adaptive dispersion is mediated by the structural category.

**Reference List**

Abramson, A. S., L-Thongkum, T., & Nye, P. W. (2004). Voice register in Suai (Kuai): an analysis of perceptual and acoustic data. *Phonetica, 61,* 147-171.

Baayen, R. H., Davidson, D. J., & Bates, D. M. (2008). Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language, 59*, 390–412.

Bailey, Guy, Tom Wikle, Jan Tillery & Lori Sand (1991). The apparent time construct. *Language Variation and Change, 3,* 241–264.

Bates, D. M., Maechler, M., & Bolker, B. (2011). Lme4 package.

Boersma, Paul & Weenink, David (2011). Praat: doing phonetics by computer [Computer program]. Version 5.2.25. <http://www.praat.org/>

Brunelle, M. (2009a). Tone perception in Northern and Southern Vietnamese. *Journal of Phonetics, 37,* 79-96.

Brunelle, M. (2009b). Contact-induced change? Register in three Cham dialects. *Journal of the Southeast Asian Linguistics Society, 2,* 1-22.

Chang, C. B. (2012). Rapid and multifaceted effects of second-language learning on first-language speech production. *Journal of Phonetics, 40,* 249-268.

Cho, H. (2011). The timing of phrase-initial tones in Seoul Korean: a weighted-constraint model. *Phonology, 28,* 293-330.

Cho, T. (1996). *Vowel correlates to consonant phonation: an acoustic-perceptual study of Korean obstruents.* University of Texas Arlington, Arlington, TX.

Cho, T. H., Jun, S. A., & Ladefoged, P. (2002). Acoustic and aerodynamic correlates of Korean stops and fricatives. *Journal of Phonetics, 30*, 193-228.

Choi, H. (2002). Acoustic cues for the korean stop contrast: dialectal variation. *ZAS papers in linguistics, 28*, 1-12.

DiCanio, C. T. (2012). Coarticulation between tone and glottal consonants in Itunyoso Trique. *Journal of Phonetics, 40,* 162-176.

Flege, J. E. (1987). The production of ''new'' and ''similar'' phones in a foreign language: Evidence for the effect of equivalence classification. *Journal of Phonetics, 15,* 47–65.

Flege, J. E., & Eefting, W. (1987). Production and perception of English stops by native Spanish speakers. *Journal of Phonetics, 15*, 67–83.

Fowler, C. A., Sramko, V., Ostry, D. J., Rowland, S. A., & Hallé, P. (2008). Cross language phonetic influences on the speech of French–English bilinguals. *Journal of Phonetics, 36,* 649-663.

Han, J.-I. (1996). *The Phonetics and Phonology of 'tense' and 'plain' consonants in Korean.*, Cornell University, Ithaca, NY.

Han, M. S., & Weitzman, R. S. (1970). Acoustic features of Korean /P, T, K/, /p, t, k/ and /ph, th, kh/. *Phonetica, 22,* 112-128.

Hardcastle, W. J. (1973). Some observations on the tense-lax distinction in initial stops in Korean. *Journal of Phonetics,* 1, 263–272.

Haudricourt, A. G. (1971). Two-way and three-way splitting of tonal systems in some far- eastern languages. In: J. G. Harris & R. B. Noss (Eds.), *Tai phonetics and phonology* (pp.58–86). Bangkok: Central Institute of English Language, Mahidol University.

Hombert, J.-M. (1978). Consonant types, vowel quality, and tone. In V. A. Fromkin (Ed.), *Tone: A linguistic survey* (pp. 77–111). New York: Academic Press.

Hombert, J.-M., Ohala, J. J., & Ewan, W. G. (1979). Phonetic explanations for the development of tones. *Language,* 55, 37-58.

Hyslop, G. (2009). Kurtöp tone: a tonogenetic case study. *Lingua, 119,* 827–845.

Jin, W. (2008). *Sounds of Chinese Korean: a Variationist Approach.* Ph.D. Thesis, Arlington, Texas: University of Texas at Arlington.

Jun, S.-A. (1993). *The phonetics and phonology of Korean prosody.* Unpublished 9401286, The Ohio State University, United States -- Ohio.

Jun, S.-A. (1996). The influence of the microprosody on the macroprosody: a case of phrase initial strengthening. . *UCLA Working paper in phonetics, 92*, 97-116.
Kagaya, R. (1974). A fiberscopic and acoustic study of the Korean stops, affricates and fricatives. *Journal of Phonetics*, 2, 161–180.
Kang, K.-H., & Guion, S. G. (2006). Phonological systems in bilinguals: Age of learning effects on the stop consonant systems of Korean–English bilinguals. *Journal of the Acoustical Society of America, 119,* 1672–1683.
Kang, K. H., & Guion, S. G. (2008). Clear speech production of Korean stops: Changing phonetic targets and enhancement strategies. *Journal of the Acoustical Society of America, 124*, 3909-3917.
Kang, Y., & Han, S. (2012). Dialectal variation in Korean stops. In *Proceedings of the 2012 joint meeting of the Society of Korean Linguistics and the Society of Korean Dialectology*.
Kessinger, R. H., & Blumstein, S. E. (1997). Effects of speaking rate on voice-onset time in Thai, French, and English. *Journal of Phonetics, 25,* 143–168.
Kim, C. W. (1965). On the Autonomy of the Tensity Feature in Stop Classification (with Special Reference to Korean Stops). *Word-Journal of the International Linguistic Association, 21*, 339-359.
Kim, M. (2004). Correlation between VOT and F0 in the perception of Korean stops and affricates. In: *Proceedings of the 8th international conference on spoken language processing (INTERSPEECH-2004)* (pp. 49–52). Jeju Island, Korea: International Speech Communication Association.
Kim, M.-R. (2000). *Segmental and tonal interactions in English and Korean: a phonetic and phonological study.* Ph.D. Thesis, Ann Arbor, Michigan: University of Michigan, Ann Arbor.
Kim, M.-R. (2008). “Lax” stops in Korean revisited: VOT neutralization. *Studies in Phonetics, Phonology and Morphology, 14,* 185-203.
Kim, M.-R. C. (1994). *Acoustic characteristics of Korean stops and perception of English stop consonants.* Ph.D. Thesis, Madison, Wisconsin: University of Wisconsin-Madison.
Kim, M.-R., Beddor, P. S., & Horrocks, J. (2002). The contribution of consonantal and vocalic information to the perception of Korean initial stops. *Journal of Phonetics, 30,* 77-100.
Kingston, J. (2011). Tonogenesis. In: M. van Oostendorp, C. J. Ewen, E. Hume, & K. Rice (Eds.), *The Blackwell Companion to Phonology*. Blackwell Publishing: Blackwell Reference Online. <http://www.companiontophonology.com/subscriber/tocnode?id=g9781405184236_chunk_g978140518423699>
Kong, E. J., Beckman, M. E., & Edwards, J. (2011). Why are Korean tense stops acquired so early?: The role of acoustic properties. *Journal of Phonetics, 39,* 196–211.
Labov, W. (1990). The intersection of sex and social class in the course of linguistic change. *Language Variation and Change, 2,* 205-254.
Ladefoged, P. (2003). *Phonetic data analysis : an introduction to fieldwork and instrumental techniques*. Malden, MA, USA: Blackwell Pub.
Lee, H. & Jongman, A. (2011). Perception of initial stops in tonal and non-tonal Korean. Poster presented at *the 162nd Meeting of the Acoustical Society of America.*
Lee, H. & Jongman, A. (in press). Effects of tone on the three-way laryngeal distinction in Korean: An acoustic and aerodynamic comparison of the Seoul and South Kyungsang dialects. *Journal of the International Phonetic Association.*
Lindblom, B. (1990). Explaining Phonetic Variation - a Sketch of the H-and-H Theory. *Speech Production and Speech Modelling, 55*, 403-439.
Lisker, L., & Abramson, A. S. (1964). A cross-language study of voicing in initial stop: acoustic measurements. *Word, 20,* 384-422.

Maran. (1973). On becoming a tone language: A Tibeto-Burman model of tonogenesis. In L. M. Hyman (Ed.), *Consonant Types and Tone, Southern California Occasional Papers in Linguistics.* (Vol. 1, pp. 98-114).

Matisoff, J. A. (1973). Tonogenesis in Southeast Asia. In Hyman, Larry (ed.), *Consonant Types and Tone, Southern California Occasional Papers in Linguistics* 1:73-95.

Mazaudon, M. & Michaud, A. (2008). Tonal contrasts and initial consonants: a case study of Tamang, a 'missing link' in tonogenesis. *Phonetica, 65*, 231-256.

Nielsen, K. (2011). Specificity and abstractness of VOT imitation. *Journal of Phonetics, 39,* 132-142.

Nützel, D. & Salmons, J. (2011). Language contact and new dialect formation: evidence from German in North America. *Language and Linguistics Compass, 5*, 705–717.

Oh, E. (2009). Voice onset time of Korean stops as a function of speaking rate. *Phonetics and Speech Sciences, 1,* 39–48. (in Korean).

Oh, E. (2011). Effects of speaker gender on voice onset time in Korean stops. *Journal of Phonetics, 39,* 59-67.

Pearce, M. (2009). Kera tone and voicing interaction. *Lingua, 119,* 846-864.

Pinheiro, J. C. & Bates, D. M. (2000). *Mixed-effects models in S and S-PLUS.* New York: Springer.

Port, R. F. (1981). Linguistic timing factors in combination. *Journal of the Acoustical Society of America, 69*, 262-274

Port, R. F. & Rottuno, R. (1979). Relation between voice-onset time and vowel duration, *Journal of the Acoustical Society of America, 66*, 654-662

R Development Core Team. (2011). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. < http://www.r-project.org/>

Sancier, M., & Fowler, C. A. (1997). Gestural drift in a bilingual speaker of Brazilian Portuguese and English. *Journal of Phonetics*, 25, 421-436.

Silva, D. J. (1992). *The phonetics and phonology of stop lenition in Korean.*, Cornell University, Ithaca, NY.

Silva, D. J. (2006). Acoustic evidence for the emergence of tonal contrast in contemporary Korean. *Phonology, 23*, 287-308.

Sohn, H.-M. (1999). *The Korean Language.* Cambridge, UK: Cambridge University Press.

Svantesson, J.-O. & House, D. (2006). Tone production, tone perception and Kammu tonogenesis. *Phonology, 23,* 309-333.

The National Institute of the Korean Language. (2005). *A Speech Corpus of Reading-Style Standard Korean* [DVDs]. Seoul, Korea: The National Institute of the Korean Language.

Thurgood, G. (2002). Vietnamese and tonogenesis. *Diachronica*, *19,* 333-363.

Whalen, D. H., & Levitt, A. G. (1995). The universality of intrinsic F0 of vowels. *Journal of Phonetics, 23,* 349-366.

Wright, J. D. (2007). *Laryngeal Contrast in Seoul Korean.* Ph.D. Thesis, Philadelphia, Pennsylvania: University of Pennsylvania.